customers_table

cust_id	cust_name
1	Rofh
2	Lerato
3	Mbali

orders_table

orde_id	cust_id	amou
101	1	500
102	2	300

Write an SQL query to give me a list of all customers and what they ordered

```
SELECT  cust_id,
        cus_name
        amount          ← LEFT table
FROM    customers table AS A
        LEFT JOIN orders_table
        AS B ON A.cust_id
        = B.cus_id
```

cust_id	cust_name	amount	order_id
1	Rofh,	500	101
2	Lerato	300	102
3	Mbali	NULL	NULL

* NULL - empty space
- missing value

Age Buckets (Age Ranges)

check the youngest person

```
SELECT MIN(Age) As min_age
       MAX(Age) As max_age
       Ave(Age) As mean_age
FROM workspace.dataset
```

are around this age

oldest person

Swift spot

```
SELECT
  CASE
    When Age = 0 Then 'infact'
    When BETWEEN 1 AND 12 THEN 'KIDS'
WHEN AGE BETWEEN 18 AND 17 THEN 'Youth'
WHEN BETWEEN 18 AND 35 'YOUTH ADULT'
WHEN BETWEEN 35 AND 50 'ADULT'
WHEN AGE > 50 AND AGE <= 60 THEN 'ELDER'
WHEN >60 THEN 'PENSIONER'
END AS Age_Group
From workspace
```

Data Types

Viewership

String: or object These are data types combination of characters STR words + num 9-2, /, >, ?; 0-9 #

Interger: number without a decimal INT eg. 8, 5

Date: A periodic also an integer

DateTime :

Timestamp : date + time (RecordDate2 in viewership

Date functions Help us to deal with dates

↳ Help us to manipulate the date columns (Process, extract, deal,

convert a string into a date YYYY/MM/DD

```
SELECT
    RecordDate2,
    TO_Date(RecordDate2) As Watch date
    DAYNAME(TO DATE(
```

TO_DATE: Converts something (a string) that is not a date to a date
YYYY-MM-DD

DAYNAME: Extracts the day name

MONTHNAME: Extracts the month name

YEAR: Extracts the year

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NULL functions

- Null is an empty space
- Null is a blank

Sales

Produc	Amount

COALESCE (col1, col2, col3, col4, ... col2)

(D in a way to replace null values by a value from alternative column or a specified value.
(if the null value doesn't work go to the next one)

We commonly use it in our SELECT Statement

Customers

cust_id	cust_name	phone_numb	whatsapp number
101	John	071 111 1111	null
102	Mary	null	082 2222
103	Peter	null	null
104	Abeding	0722448	0825421

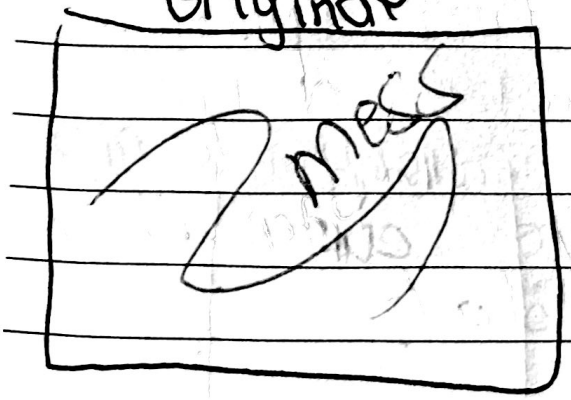
write an SQL query to give the name of the customer, and their contact details

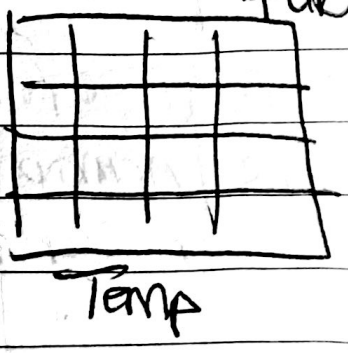
```
SELECT cust_name,
       COALESCE (phone_number,
       whatapp_num, 0000)
       AS contact_number
FROM customers;
```

cust_name	contact_name
John	0711111111

- Summarized version of a bigger
- Creating a snippet based on another table

Temporary table
orginal





eg: movie trailer

Joins

- We want to combine multiple tables into one single table
- We "Join" tables based on a common column between the different tables.

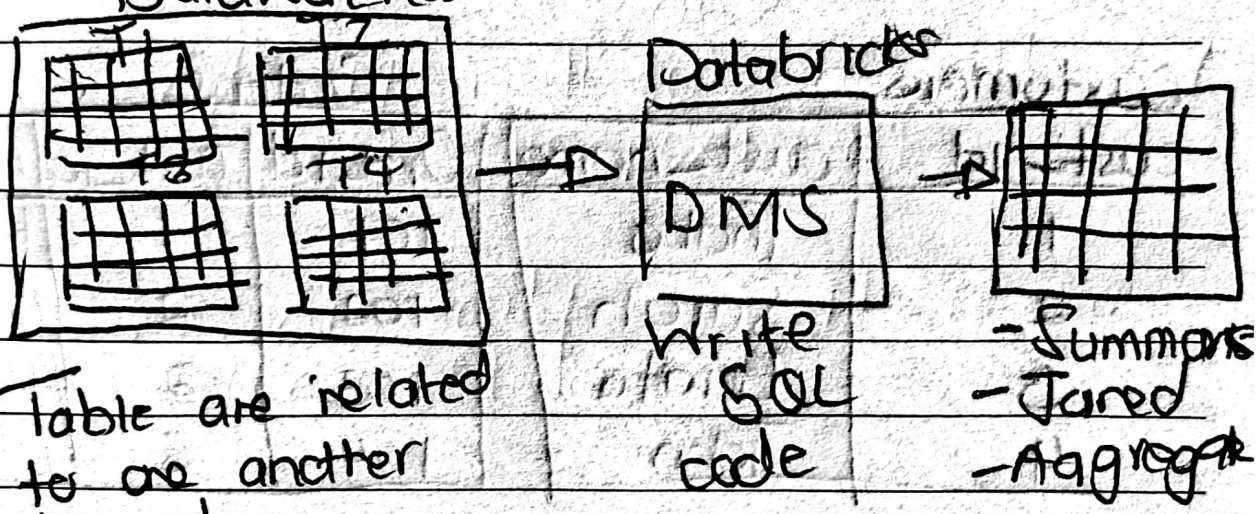


Table are related to one another based on a Foreign key or a primary key

How we Join Tables

INNER JOINS

- Joins tables, and returns only rows that have matching values on both tables

```
SELECT col1, col2, col3
FROM table1 AS A
INNER JOIN table2 AS B
ON A.common_column = B.
   common_column;
```

Customers

cust_id	cust_name
1	Thabo
2	Sarah
3	Lerato
4	John

Orders

Ord_id	cust_id	Produ
101	1	Lapto
102	2	Phone
103	2	Headph
104	5	Tablet

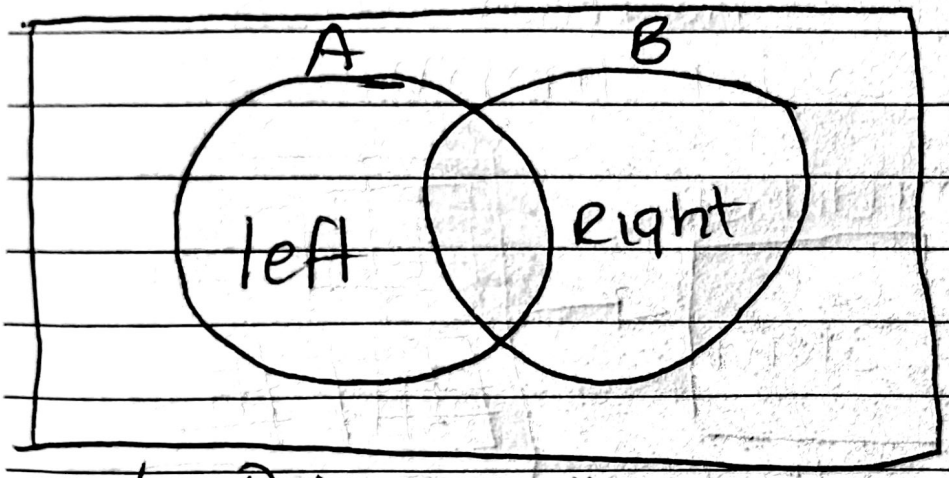
```
SELECT
    CustomerID,
    customerName,
    OrderID,
    Product
FROM customers AS A
INNER JOIN orders AS B
ON A.customerID = B.customerID;
```

CustomerID	CustomerName	OrderID	Product
1	Thabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Headphon

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A.cust

LEFT JOIN



↳ Returns all the rows from the left table and only the matching rows from the right table

```
SELECT col1,
       col2,
       col3
From table AS A
LEFT JOIN table AS B
ON A.common.column = B.common.colum
```

Customer

Charts

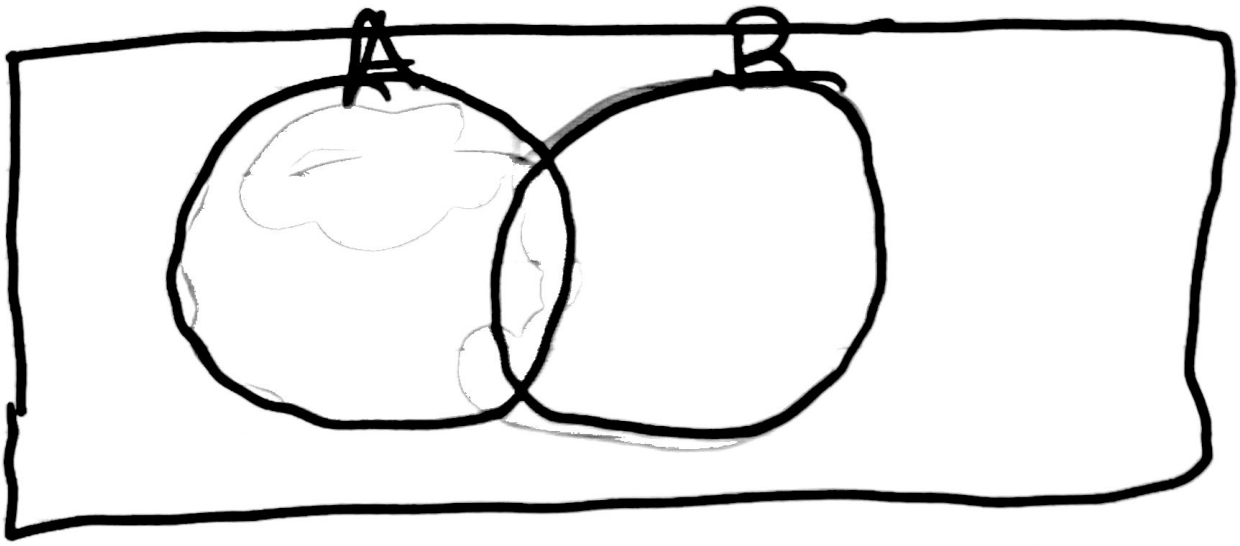
Chart type	Uses	Avoid
Bar Column	comparing categ or time period	ist categorie
line	Showing trents over time	Data points are not in a trend region
Pie / Donut	Shou breakdown of a whole	Not more than 6 categ
Scatteplot	Finding corelation belwe 2 variab	Your audien is non technic
Histogram	Showing data distribution / freq	You have fewer tha 20 dat points
waterfal	Explaing step by step	Changes are not in seq
Tree map	Best for showing how parts make up a whole	Categories are of equal importance

Full Outer Join

A FULL OUTER JOIN returns

- All matching rows
  - All the rows that are matching on a common column are combined into a single row
- All unmatched row
  - where there's no match we will put a NULL

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```
SELECT col1, col2, cols
FROM table1 As A
FULL OUTER JOIN table2 AS B
ON A.common_column = B.common_column
```

Customers

customer ID	Customer_Name
1	Thabo
2	Sarah
3	Lerato
4	John

orders

Order ID	Custer ID	Produ
101	1	Laptop
102	2	Phone
103	2	Headphone
104	5	Tablet

```
SELECT custome ID
Customer Name
Order ID
product
```

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```
From customers AS C
FULL OUTER JOIN orders AS O
ON C.customersID = O.customerID;
```

Customer ID	Customer Name	order ID	Product
1	Thabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Headphones
3	Lerato	NULL	NULL
4	John	NULL	NULL
6	NULL	104	NULL

UNION & UNION ALL

- Combine the results of two SELECT statements by stacking the rows
- The Select Statements must have the same number of columns
- The columns must have the same data types
- Remove duplicates
- does not remove duplicates

Syntax:

order must match

difference: table names

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Current students

Student_id	Student Name
1	Thabo
2	Sarah
3	Lerato

New students

Student_id	Student_Name
3	Lerato
4	John
5	Mary

UNION

```
SELECT  Student_id
        StudentName
FROM    current_students
 UNION
SELECT  Student_id
        StudentName
FROM    New_students
```

Student-id	Student Name
1	Thabo
2	Sarah
3	Lerato
4	John
5	Mary

UNION ALL

```
SELECT  Student_id
        StudentName
FROM    current_students
 UNION  ALL
SELECT  Student_id
        StudentName
FROM    New_students
```

Student-id	Student Name
1	Thabo
2	Sarah
3	Lerato
4	John
5	Mary
3	Lerato

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